

## Title

# Regional Anesthesia Techniques for Cesarean Delivery: Comparative Efficacy and Postpartum Mental Health Outcomes

## Abstract

Adequate postoperative analgesia following cesarean section is essential for maternal recovery and may influence postpartum mental health. Two regional anesthesia approaches are commonly used, yet comparative effectiveness for pain control and mental health outcomes remains unclear. This randomized study will compare two distinct regional anesthesia blocks in women undergoing cesarean delivery. Sixty postoperative women will be randomized 1:1 to receive either an erector spinae plane block or a transversus abdominis plane block following their surgical procedure. Pain severity will be assessed using standardized visual analog scales at multiple timepoints postoperatively. Secondary outcomes include requirement for rescue analgesics, patient satisfaction with pain management, and postpartum depression screening using standardized questionnaires at 4-6 weeks postpartum. All participants will receive standard postoperative care protocols. This comparative effectiveness study will provide data on whether the choice of regional anesthesia technique influences both acute pain outcomes and postpartum mood.